Photonic Time Crystals in Moiré Superlattices

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Abstract

Opening a momentum bandgap in a photonic time crystal requires a modulation depth that is prohibitively large at optical frequencies. Although resonances have been shown to lower this threshold by enhancing light–matter interaction, band flatness and radiative lifetime remain intertwined, restricting independent control of the dispersion. Here we demonstrate that moiré superlattices overcome this limitation, as the long-wavelength moiré potential determines the miniband bandwidth through structural design rather than modal lifetime. A tight-binding model yields a closed-form condition for full-zone amplification, governed by the interplay between the modulation-induced coupling, the residual bandwidth, and the decay rate. We show that a moiré flat band opens a momentum bandgap across the entire Brillouin zone at a modulation depth far below the bulk requirement, whereas a dispersive miniband of the same structure necessitates orders of magnitude stronger modulation despite its longer radiative lifetime. The same condition extends to an isolated moiré supercell, where the threshold is set solely by the decay rate in the absence of miniband dispersion, recovering the condition for parametric resonance in an open resonator. These results connect moiré physics with time-varying photonics, opening a route toward new parametric phenomena in structured media.

Keywords— photonic time crystals, moiré superlattices, photonic flat bands, momentum bandgaps, Floquet physics

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1 Introduction

Breaking time-translation symmetry underlies effects as distant as parametric resonance in driven systems, particle creation in expanding spacetimes, and the dynamical Casimir effect [1–3]. This, in turn, gives rise to a broad range of light–matter interaction phenomena, including photon acceleration, temporal reflection, synthetic gauge fields, nonreciprocal transport, dynamic quantum-state control, directional scattering, and parametric amplification [4–14]. Photonic time crystals (PTCs) constitute a notable example. Temporal periodicity replaces the spatial variations of a conventional photonic crystal, so that momentum is conserved while energy can be exchanged with the dynamic matter. A momentum bandgap therefore emerges, within which the eigenfrequencies become complex and the corresponding modes grow exponentially in time. This can be envisioned across a range of physical regimes, in which the parametric process enables amplified spontaneous emission, free-electron light emission, and the coherent amplification of incident fields [15–20].

Bulk realizations of PTCs require ultrafast modulation accompanied by a modulation depth on the order of unity [21]. At optical frequencies, various approaches have been implemented to induce changes in the material response within a single optical cycle, most notably by pumping the epsilon-near-zero (ENZ) regime of indium tin oxide and aluminium-doped zinc oxide, where large permittivity variations allow for temporal refraction [6, 22, 23]. Nevertheless, opening a momentum bandgap requires the modulation to persist periodically over several cycles, a condition that has not yet been realized in these materials, largely because of the associated energy constraints [24]. Material dissipation also remains a major limiting factor, as the modulation must not only open a momentum bandgap but also overcome absorption to produce net amplification [25]. In this context, the optical Kerr nonlinearity of transparent dielectrics such as silicon and germanium offers a pathway toward effectively instantaneous modulation with low absorption [26–28]. The nonlinear response, however, remains inherently weak, leaving the attainable modulation depth orders of magnitude below that required for a bulk PTC [21, 26].

Spatial structuring offers a route to circumvent this limitation by introducing optical resonances that increase the interaction time between the field and the modulated medium. Prolonged light–matter interaction therefore enables even a weak temporal modulation to generate appreciable parametric coupling. Exponential amplification and full momentum bandgaps have been demonstrated experimentally in the microwave regime [29, 30]. In the optical range, Mie-resonant metasurfaces have been theoretically shown to substantially enlarge the momentum bandgap under weak modulation, while nanoresonators supporting quasi-bound states in the continuum (qBIC) further reduce the required threshold by suppressing radiative leakage [25, 31]. The flatness of the underlying band determines whether the gap can span the entire Brillouin zone. Relying on qBICs of individual meta-atoms, however, intertwines reduced radiative leakage with flatness, as both arise from strong field confinement within the constituent resonators. Achieving a full momentum bandgap at optical frequencies therefore remains a key challenge, as band flatness and radiative confinement remain closely tied in these systems.

Moiré structures constitute a distinct platform in which the band dispersion is controlled through geometric parameters. They are formed by the superposition of periodic lattices with slightly mismatched periods or orientations, generating a long-wavelength potential that reconstructs the underlying bands. In electronic systems, the moiré potential can suppress the effective kinetic energy and drive the formation of nearly flat bands, as observed in twisted bilayer graphene near the magic angle [32–34]. Analogously, photonic moiré superlattices consisting of mismatched photonic crystal slabs can support minibands whose dispersion is governed by the interplay between intralayer diffraction and interlayer evanescent coupling [35, 36]. The miniband can consequently be driven into an almost dispersionless regime through geometric tuning of the structure. At the same time, moiré flat bands can retain long radiative lifetimes, resulting in optical modes with high quality factors (Q-factors) [37–39]. The miniband bandwidth and radiative leakage can be traced back to distinct physical mechanisms, with the former governed by the interplay of intra- and interlayer coupling and the latter by radiative coupling to the continuum [35, 38]. Together, these properties make moiré superlattices a promising platform for opening momentum bandgaps under weak temporal modulation.

Here we investigate PTCs in bilayer moiré superlattices, where geometric control of the miniband dispersion allows the temporal modulation to act on a flat photonic band (Figure 1). Within a tight-binding description of the miniband, we derive a closed-form amplification condition governed by the modulation-induced coupling, residual bandwidth, and decay rate. The resulting criterion establishes the residual miniband dispersion as the dominant scale at low loss, with dissipation taking over only once the decay rate approaches the bandwidth. Under realistic material loss, our analysis demonstrates that the moiré potential enables a full momentum bandgap at pump intensities well below the damage threshold. A more dispersive miniband in the same structure, however, requires over two orders of magnitude stronger modulation, despite exhibiting lower radiative loss. The same framework also extends to the isolated-supercell limit. Because the flat-band mode is confined within a single cell, the lattice can be truncated, and the criterion reduces to the parametric-resonance condition of an open time-varying resonator.

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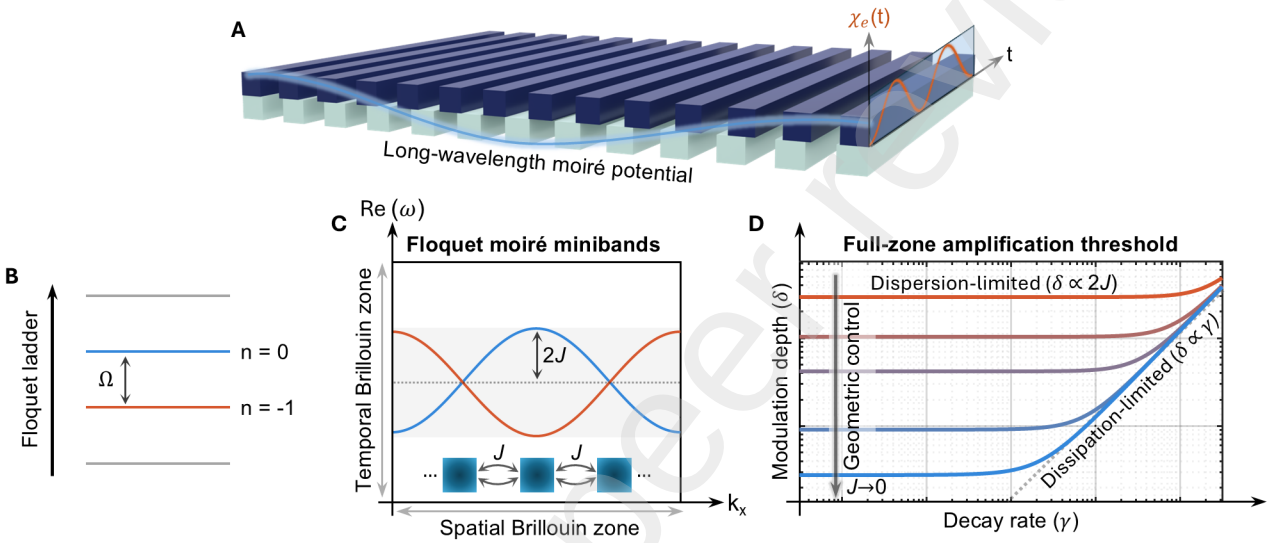


Figure 1: **Concept of moiré-enabled photonic time crystals.** (A) Temporally modulated bilayer grating forming a moiré superlattice. The period mismatch produces a long-wavelength moiré potential that can be tuned geometrically to control the miniband dispersion; the susceptibility modulation $\chi_e(t)$ of the upper grating is shown in the inset. (B) Floquet ladder showing the resonantly coupled $n = 0$ and $n = -1$ harmonics, separated by the modulation frequency Ω and degenerate at the edge of the temporal Brillouin zone. (C) Floquet moiré minibands crossing at the edge of the temporal Brillouin zone, where the modulation couples them. The inset shows the tight-binding picture of the superlattice, in which the nearest-neighbor hopping rate J sets the residual bandwidth $4J$. (D) Threshold modulation depth for full-zone amplification, shown as a function of the decay rate for progressively flatter moiré minibands obtained through geometric tuning. Each curve is flat while the decay rate stays below the miniband bandwidth, with a plateau set by $2J$, and rises once dissipation takes over. All curves converge onto a common asymptote at large decay rate, where the advantage of a flat band is lost.

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2 Results

2.1 Photonic flat bands in a moiré superlattice

We consider a bilayer dielectric structure composed of two one-dimensional photonic crystal slabs, as illustrated in Figure 2A. The two slabs are taken to have a refractive index $n_g = 3.982$, corresponding to germanium near a wavelength of $4\,\mu$m [40]. The mismatch between the two layers is introduced through slightly different lattice periods, Λ_1 and Λ_2, subject to the commensurate condition $(N-1)\Lambda_1 = N\Lambda_2 = \Lambda$, where Λ denotes the period of the resulting moiré superlattice. Here, N is a positive integer that sets the commensurate relation between the two constituent lattices, such that one moiré supercell contains N periods of the upper layer and $N-1$ periods of the lower layer. Each slab is characterized by a ridge width w_j and thickness t_j. The two layers share the same filling factor f_w and thickness fraction f_t. With the average lattice period defined as $\Lambda_0 = (\Lambda_1 + \Lambda_2)/2$, the ridge widths satisfy $w_1 = f_w\Lambda_1$ and $w_2 = f_w\Lambda_2$, whereas both slabs have the same thickness, $t_1 = t_2 = f_t\Lambda_0$. The period mismatch brings about the relative alignment of the two gratings to vary across each moiré supercell. This produces locally registered AA regions, where the ridges of the two layers are aligned, and AB regions, where the relative displacement between the two layers is largest (see Figure 2A). This arrangement, in turn, gives rise to a slowly varying moiré potential that reshapes the photonic dispersion and provides the conditions for the bilayer superlattice to support flat bands associated with highly confined optical modes.

The photonic response of the bilayer moiré superlattice can be understood in terms of the coupling among co- and counter-propagating Bloch guided modes supported by the two photonic crystal slabs [35]. Within each slab, counter-propagating guided modes are coupled through diffraction near the corresponding high-symmetry points of the Brillouin zone. Co-propagating modes supported by the two layers, meanwhile, interact through evanescent coupling. The interplay between these mechanisms governs the formation and dispersion of the moiré minibands. Herein, the two gratings are in direct contact, and the slab thicknesses are held fixed, so that the evanescent coupling remains essentially constant while the in-plane geometry is varied. We then tune the filling factor, which directly modifies the intralayer diffractive coupling within each slab and brings the miniband dispersion close to the flat-band limit. We demonstrate below that the resulting flat-band states are tightly confined and, for an appropriate choice of geometry, exhibit high quality factors associated with strongly suppressed radiative leakage.

We next examine the photonic band structure as the filling factor is varied, with the remaining geometric parameters kept constant. The two gratings are placed in direct contact, and the thickness fraction is fixed at $f_t = 0.3$. This particular choice not only maintains strong interlayer coupling but also preserves tight confinement of the guided modes. Throughout, we consider TE-polarized (s-polarized) modes, for which the electric field is oriented along the invariant direction of the gratings (y-axis in our geometry). This polarization supports tightly bound guided modes in the individual slabs. Finally, we take $N = 14$ in our calculations, which corresponds to a representative commensurate supercell without affecting the qualitative behavior. Figure 2B–D presents the photonic band structures for filling factors $f_w = 0.8$, 0.6, and 0.4. The modes are identified from the minima of $\sigma_{\min}$, the smallest singular value of the inverse Floquet–Bloch scattering matrix, plotted on a logarithmic scale [41] (see Methods). As the filling factor is reduced from $f_w = 0.8$ to 0.4, the target band (highlighted in Figure 2B–D) gradually evolves into a flat band, with its dispersion becoming nearly independent of k_x across the entire moiré Brillouin zone. Over the same range, the band shifts toward higher frequencies, in line with the reduced average refractive index of the gratings. Concurrently, the dispersive manifold of folded guided modes retreats to lower frequencies, such that at $f_w = 0.4$ the desired flat band becomes spectrally isolated from the underlying continuum. The emergence of the isolated flat band is consistent with the aforementioned interplay between intralayer and interlayer coupling mechanisms [35, 36].

Figure 2E illustrates the field profile of the target flat-band mode at the Γ point for $f_w = 0.4$. The field is strongly localized at the center of the moiré supercell, within the AB region, and resides predominantly in the top photonic crystal slab. The field amplitude decays progressively toward the AA regions near the edges of the supercell, indicating strong spatial confinement induced by the moiré potential [36, 37]. We now consider the radiative characteristics of this mode, which can be quantified by the Q-factor shown in Figure 2F. Since the moiré period greatly exceeds the primitive lattice constants, the corresponding reciprocal lattice vector is small, which in turn allows for several diffraction orders to fall within the radiative region across the Brillouin zone [38]. For the flat band considered here, the lowest-order diffraction channels remain open throughout the Brillouin zone, whereas the next pair lies just above its Rayleigh anomaly and becomes radiative only within a narrow interval around normal incidence (indicated by the dashed lines in Figure 2F). These additional radiative channels lower Q near the Γ point. As they close, Q recovers and increases toward the edges of the moiré Brillouin zone. Overall, Q remains on the order of 10^6 across the entire zone, indicating that the mode combines tight spatial confinement with negligible radiative loss.

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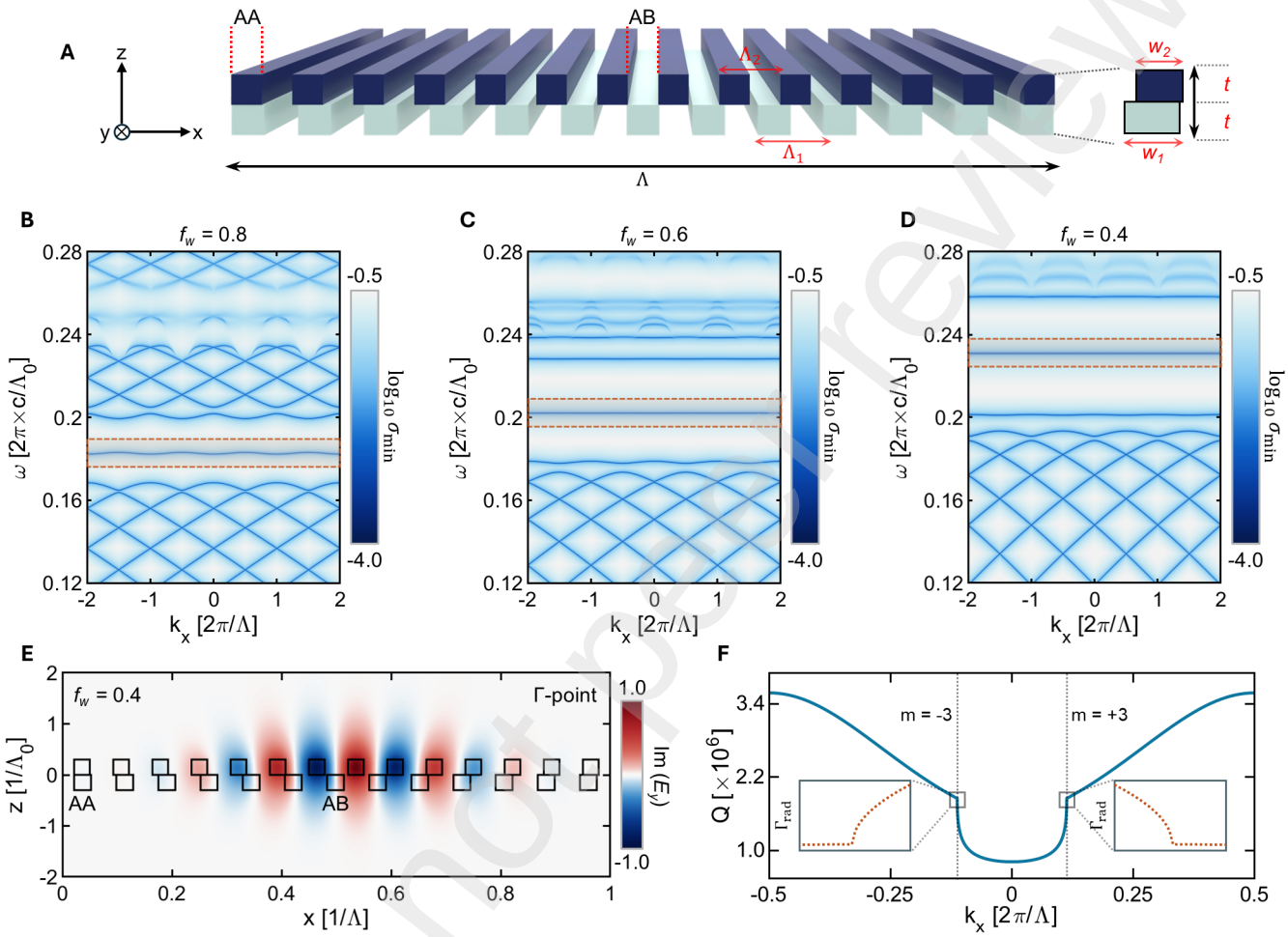


Figure 2: **Photonic flat bands in a bilayer moiré superlattice.** (A) Geometry of the commensurate bilayer superlattice, showing the AA and AB stacking regions. (B–D) Photonic band structures for filling factors $f_w = 0.8$, 0.6, and 0.4. The target miniband flattens progressively and shifts to higher frequency as f_w is reduced, becoming spectrally isolated from the underlying dispersive manifold. Bands are obtained from the minima of $\sigma_{\min}$, the smallest singular value of the inverse Floquet–Bloch scattering matrix. (E) Electric-field profile of the flat-band mode at the Γ point for $f_w = 0.4$, concentrated at the AB region in the center of the moiré supercell and residing predominantly in the upper slab. (F) Quality factor across the moiré Brillouin zone. The reduction near the Γ point arises from the $m = \pm 3$ diffraction orders, which are radiative only within the interval marked by the dashed lines.

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2.2 PTCs enabled by moiré flat bands

As described above, the interplay of intralayer and interlayer coupling mechanisms in the moiré superlattice brings about a flat band with vanishing group velocity, strong field confinement, and negligible radiative loss. As the band remains nearly resonant across the entire Brillouin zone, merely a perturbative temporal modulation is required to couple the associated Floquet states over a broad range of momenta. This makes the moiré superlattice an appealing platform for the study of PTCs. Since the flat-band modes are strongly localized within individual moiré supercells, the system can be viewed, to leading order, as a chain of weakly coupled sites, with each localized mode playing a role analogous to an atomic orbital in a crystalline lattice [42] (see Figure 3A). Accordingly, the resulting miniband can be described, to leading order, by a nearest-neighbor tight-binding model [35, 39]:

$$\omega_k = \omega_0 + 2J\cos(k_x\Lambda). \tag{1}$$

Here, ω_k denotes the resonance frequency of the flat band at wavevector k_x, ω_0 the miniband center frequency, and J the nearest-neighbor hopping rate quantifying the coupling between adjacent supercells. The total bandwidth is $4J$, so that a flat band is associated directly with a small hopping rate. The parameters ω_0 and J are extracted by fitting Equation 1 to the computed miniband dispersion. This tight-binding description is particularly convenient for describing the PTC, since the miniband flatness is captured by the single parameter J, which can be compared directly with the strength of the temporal modulation.

We temporally modulate solely the susceptibility of the top grating, taking advantage of the fact that the flat-band mode is predominantly localized within the upper photonic crystal slab (see Figure 2E). As a result, the mode experiences nearly the full modulation even when a single layer is driven, as illustrated in Figure 3B. The time-dependent susceptibility is taken as

$$\chi_e(t) = \chi_0\left[1 + \delta\cos(\Omega t)\right], \tag{2}$$

where $\chi_0 = n_g^2 - 1$ denotes the static susceptibility, δ the modulation depth, and Ω the modulation frequency. We now examine the momentum bandgaps induced by the temporal modulation. Of particular interest is how their formation is affected by the flatness of the underlying moiré miniband. To this end, we consider the dispersive band at $f_w = 0.8$ and the flat band at $f_w = 0.4$. In both scenarios, the Q-factor is on the order of 10^6 or higher. Radiative loss therefore remains negligible, allowing the effect of miniband bandwidth to be isolated effectively.

The static minibands and their corresponding tight-binding fits are shown in Figures 3C and 3F. The extracted hopping rates are $J \approx 2.04\times10^{-4}$ for the dispersive band and $J \approx 6.63\times10^{-7}$ for the flat band. Since the bandwidth is defined as $\Delta\omega = 4J$, the latter is narrower by more than two orders of magnitude. The temporal modulation folds the spectrum into the first temporal Brillouin zone. The relevant Floquet branches therefore become degenerate at its edge, $\mathrm{Re}(\omega) = \Omega/2$. We place this degeneracy at the Γ-point resonance for the dispersive band, $\Omega/2 = \omega_0 + 2J$, and at the miniband center for the flat band, $\Omega/2 = \omega_0$, which keeps the detuning small across the Brillouin zone. We drive both cases at the same weak modulation depth, $\delta = 3.4\times10^{-5}$, corresponding to a susceptibility variation of only 0.003%. In both cases, a momentum bandgap opens at $\mathrm{Re}(\omega) = \Omega/2$, as shown in Figures 3D and 3G. For the dispersive band, the gap is confined to a narrow interval around $k_x = 0$ and is barely resolved at the scale of the panel (inset of Figure 3D), as the miniband rapidly detunes from resonance away from the Γ point. For the flat band, by contrast, the gap extends across the entire moiré Brillouin zone. Within the momentum bandgap, the Floquet eigenfrequencies become complex, with their imaginary parts shown in Figures 3E and 3H. The branch with positive imaginary part corresponds to temporal amplification. For the dispersive band, this instability is confined to only $\Delta k \approx 2\%$ of the moiré Brillouin zone, whereas for the flat band it spans the full momentum range. Both minibands support high-Q modes, yet only the flat one develops a momentum bandgap across the full Brillouin zone. Weak radiative loss is therefore not sufficient on its own, and the miniband bandwidth, set independently by the moiré geometry, emerges as a key parameter governing the PTC.

To shed light on the physics underlying the momentum bandgap in the moiré superlattice, we reduce the driven miniband to a minimal two-mode Floquet model describing the coupling between the two degenerate branches at the temporal Brillouin zone edge. Specifically, we retain the two degenerate Floquet harmonics, $n = 0$ and $n = -1$, which are resonantly coupled by the temporal modulation. The dynamics of these two resonant Floquet harmonics can then be described by the effective Hamiltonian [43]

$$H_k = \begin{bmatrix} \Delta_k - i\gamma_k & g \\ -g & -\Delta_k - i\gamma_k \end{bmatrix}, \qquad \Delta_k = \omega_k - \frac{\Omega}{2}. \tag{3}$$

Here, Δ_k is the detuning from the temporal Brillouin-zone edge, γ_k the total decay rate accounting for both radiative and material losses, and g the modulation-induced coupling strength (see Section S1 of the Supporting Information

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for further details). In the perturbative regime considered here, $g = C\delta\Omega/2$, with C a dimensionless coefficient set by the overlap of the mode with the modulated region [44]. The corresponding quasifrequencies follow from the eigenvalues of H_k as

$$\tilde{\omega}_{\pm}(k_x) = \frac{\Omega}{2} - i\gamma_k \pm \sqrt{\Delta_k^2 - g^2}. \tag{4}$$

A momentum bandgap therefore requires $|\Delta_k| < g$. Using the tight-binding description of Equation 1 and setting $\Omega = 2\omega_0$, the detuning becomes $\Delta_k = 2J\cos(k_x\Lambda)$. Hence, $g > 2|J|$ is necessary and sufficient for the momentum bandgap to span the entire Brillouin zone. Nevertheless, amplification requires more than the opening of a bandgap, since the growth rate must also exceed the total decay rate,

$$\sqrt{g^2 - \Delta_k^2} > \gamma_k. \tag{5}$$

Equation 5 gives the general condition for amplification across the Brillouin zone. Here we neglect material absorption, so γ_k represents only radiative decay. For the moiré flat band, this lies more than an order of magnitude below g, so the amplification condition collapses onto the bandgap condition $g > 2|J|$ obtained above. With $C \approx 0.2$ for the structure considered here, full-zone amplification requires $\delta \gtrsim 2.9 \times 10^{-5}$, slightly below the modulation depth $\delta = 3.4 \times 10^{-5}$ used above. Applying the same criterion to the dispersive band gives $\delta \gtrsim 8.9 \times 10^{-3}$, over 300-fold larger. The distinction between the two cases is therefore set by the hopping rate J, which can be tuned by manipulating the moiré potential through a single geometrical parameter.

2.3 Material loss and finite moiré cavities

The analysis so far has established band flatness as a critical element for realizing a PTC across the entire Brillouin zone. We now move beyond the idealized assumptions of a lossless material and an infinitely extended moiré superlattice. First, we investigate how material absorption modifies the amplification threshold. We then determine whether the same parametric response persists in a finite moiré structure. Including loss, however, requires a causal description of the dispersive time-varying medium, as temporal modulation couples Floquet harmonics that sample different regions of the material dispersion. The susceptibility must therefore satisfy the Kramers–Kronig relations [41,45]. To account for this behavior, we represent the optical response of germanium using a Drude–Lorentz model fitted to experimental data [40] (additional details are provided in Section S3 of the Supporting Information). Dispersion also breaks the scale invariance of the problem, so an absolute frequency must be specified. Hence, we choose $\Lambda_0 = 923.6$ nm, which sets the flat band at a free-space wavelength of $4\,\mu$m, well within the transparency window of germanium.

Figure 4A shows the threshold modulation depth, $\delta_{\min}$, required to sustain amplification across the entire Brillouin zone as a function of material loss. This is quantified through the extinction coefficient κ, which is controlled by adjusting the damping rate in the Drude–Lorentz model while keeping the refractive index at the operating wavelength constant. The threshold falls into two distinct regimes. At low material loss, where the decay rate stays far below the hopping rate, it remains nearly constant and is governed by the residual miniband dispersion. In this regime, the modulation must overcome the detuning of the band edges from $\Omega/2$, and Equation 5 reduces to $g > 2|J|$. The quality factor falls by nearly an order of magnitude across this plateau while $\delta_{\min}$ stays almost unchanged, demonstrating that the threshold here is set by band flatness rather than by Q. By contrast, as the loss increases and γ_k becomes comparable to or larger than $2|J|$, dissipation becomes the dominant limitation and the threshold rises with κ. The proposed Hamiltonian closely reproduces the full-wave results, confirming that the two regimes are governed by the interplay of g, J, and γ_k. For germanium at $4\,\mu$m, the intrinsic absorption keeps the structure deep within the detuning-limited regime. Utilizing the measured nonlinear refractive index of germanium at this wavelength [46], the required modulation depth corresponds to a Kerr pump intensity below $1.5\,\mathrm{GW/cm^2}$, several times lower than the laser-induced damage threshold of nanostructured germanium [47, 48]. A momentum bandgap spanning the entire Brillouin zone is therefore attainable in a conventional dielectric under experimentally accessible pumping conditions.

As mentioned earlier, the tight-binding model assumes that the flat-band mode is localized within an individual moiré supercell, with only weak coupling to neighboring cells. Wannier functions provide a natural basis for this description [42]. The Wannier function of the moiré miniband, computed in the twisted parallel-transport gauge [49], is shown in Figure 4B (see Section S3 of the Supporting Information for additional discussion). More than 95% of the Wannier density is confined to a single supercell, which supports the nearest-neighbor description and suggests that the mode should persist even in an isolated supercell. Figure 4C shows the field profile of the resonance supported by an isolated moiré supercell terminated in free space. The mode retains the characteristic localization of the superlattice and exhibits a Q-factor of approximately 1.65×10^5. Although truncation introduces additional

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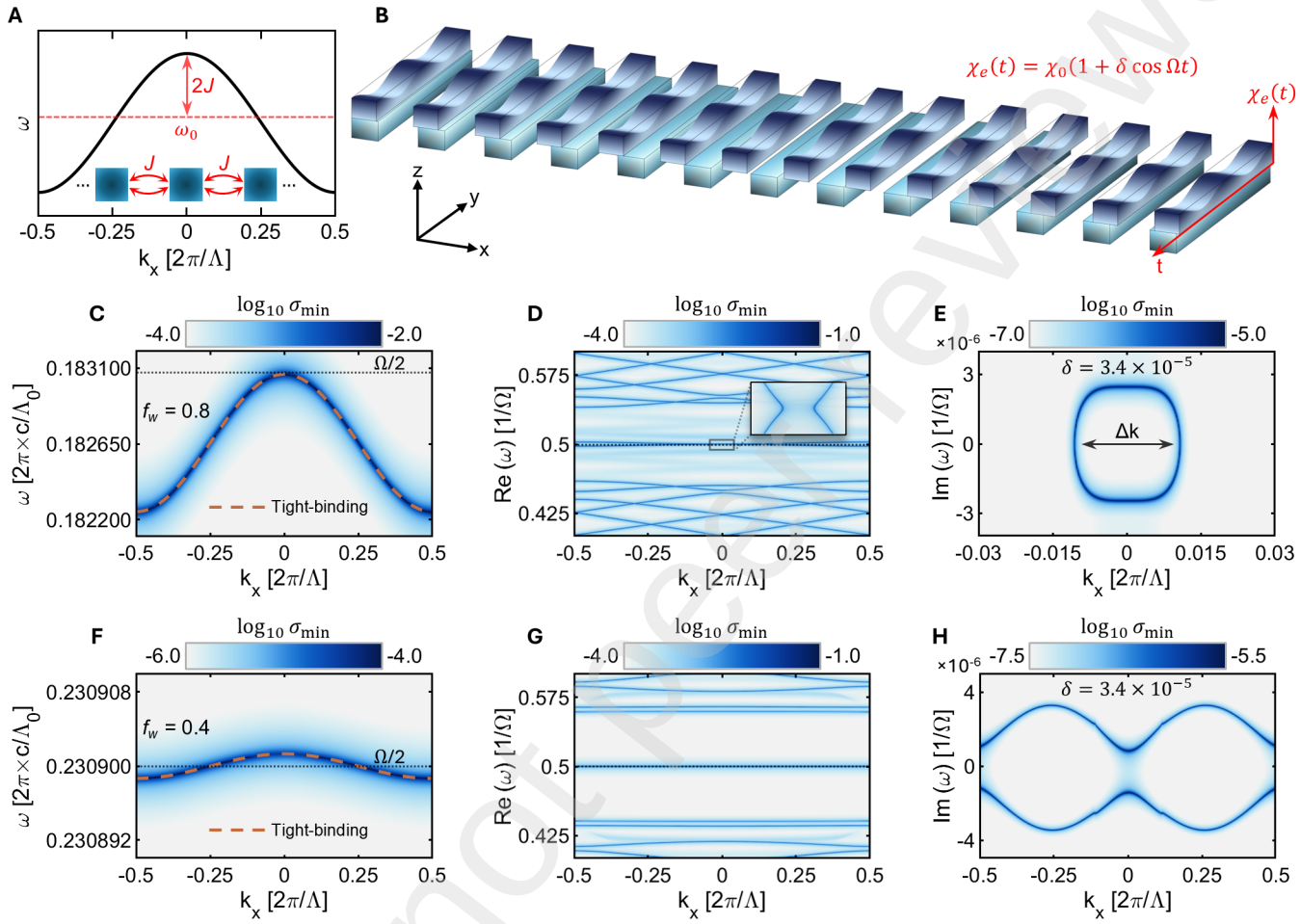


Figure 3: **Photonic time crystals enabled by moiré flat bands.** (A) Tight-binding description of the moiré miniband as a chain of supercell-localized modes coupled at a rate J, which sets the bandwidth $4J$. (B) Temporal modulation of the susceptibility applied only to the upper photonic crystal slab, exploiting the concentration of the flat-band mode within that layer. (C–E) Static dispersion with the tight-binding fit, real part of the quasifrequency, and growth rate under temporal modulation for the dispersive miniband at $f_w = 0.8$, with $J = 2.04 \times 10^{-4}$ in units of $2\pi c/\Lambda_0$. (F–H) Corresponding quantities for the flat miniband at $f_w = 0.4$, with $J = 6.63 \times 10^{-7}$, more than two orders of magnitude smaller. Both configurations are driven at the same modulation depth, $\delta = 3.4 \times 10^{-5}$, and support modes with comparable quality factors. The dispersive band develops a momentum bandgap only over a narrow interval around the tuning point, whereas the flat band sustains a momentum bandgap across the entire Brillouin zone, with net amplification at every momentum.

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radiative leakage at the supercell boundaries and reduces Q by roughly an order of magnitude, the residual decay rate remains low enough to support parametric resonance, as demonstrated below. Reducing the structure to a single supercell is experimentally advantageous, since the pump can uniformly address the entire device. More importantly, the isolated resonator has no miniband dispersion, so the amplification threshold is set solely by the decay rate. This threshold as a function of κ is shown in Figure 4D. With the detuning term absent, Equation 5 reduces to $g > \gamma$, giving $\delta_{\min} = 1/(2CQ)$. For the calculated Q, this yields $\delta_{\min} \approx 1.5 \times 10^{-5}$. This value is lower than in the periodic structure even though Q is an order of magnitude smaller, since the modulation no longer has to overcome the miniband detuning. The result is consistent with the parametric resonance condition for open PTCs [44], and the full-wave calculations reproduce it across the entire range of κ. This modulation depth enables parametric amplification at a Kerr pump intensity ($< 1\,\mathrm{GW/cm^2}$), an order of magnitude below the damage threshold of nanostructured germanium [47, 48]. Amplification of the moiré flat-band mode is therefore sustained in a single finite supercell at a lower modulation threshold than in the extended lattice.

3 Discussion

We have investigated PTCs in bilayer moiré superlattices, where the interplay of intra- and interlayer coupling enables the miniband bandwidth to be tuned through a single geometric parameter. Our findings establish that a low radiative decay rate is not, by itself, sufficient to open a momentum bandgap spanning the entire Brillouin zone. Once losses are sufficiently small, the required modulation depth is governed by the flatness of the miniband. The moiré potential can be tuned to suppress the residual dispersion while preserving negligible radiative loss, enabling a full momentum bandgap at a modulation depth as low as $\delta \approx 2.9 \times 10^{-5}$. In contrast, the miniband with more pronounced dispersion, despite its longer radiative lifetime, requires a modulation depth over two orders of magnitude larger. We formulate this behavior through a minimal Floquet Hamiltonian and derive analytical amplification conditions in terms of the moiré miniband parameters, in excellent agreement with the full-wave simulations. The threshold is largely unaffected by loss until the decay rate approaches the residual miniband bandwidth, above which dissipation sets the modulation requirement.

A Wannier analysis of the miniband shows that the flat-band mode is confined almost entirely within a single moiré supercell, indicating that the essential resonant physics is already captured at the level of an individual cell. We demonstrate that the mode persists in an isolated supercell, where truncation lowers the Q-factor but leaves it high enough to support parametric amplification. In the absence of miniband dispersion, the same Hamiltonian predicts a threshold set solely by the decay rate, and the isolated supercell reaches the parametric regime at $\delta \approx 1.5 \times 10^{-5}$, below the value required for the extended lattice despite its lower Q. This limit recovers the parametric resonance condition of an open resonator. Momentum-bandgap formation and resonant parametric amplification therefore arise as two limits of the same criterion, determined by whether dispersion or dissipation sets the dominant scale. The extended lattice nonetheless retains a distinct advantage, since a momentum bandgap spanning the entire Brillouin zone provides parametric amplification for every Bloch state, a functionality that has no direct counterpart in an isolated resonator.

For germanium in the mid-infrared, the proposed configurations reach threshold at experimentally accessible pump intensities. A momentum bandgap spanning the entire Brillouin zone requires less than $1.5\,\mathrm{GW/cm^2}$ in the periodic superlattice, while parametric amplification in an isolated supercell requires less than $1\,\mathrm{GW/cm^2}$. In both cases, the required intensity remains several times below the reported laser-induced damage threshold of nanostructured germanium. While these estimates are based on idealized structures, they indicate that the proposed mechanism lies within reach of current experimental capabilities. These operating conditions open new opportunities for classical and quantum photonics, including wide-angle optical amplification, amplified spontaneous emission, and compact parametric resonators operating at low modulation depth. The underlying physical principle is not specific to germanium or to the wavelength considered here, suggesting that moiré superlattices provide a practical route toward the realization of PTCs at optical frequencies.

Methods

Spatiotemporal modes

The electromagnetic modes of the moiré superlattice are labeled by a Bloch wavevector k_x and, under temporal modulation, by a quasifrequency ω [41]. Both quantities are restricted to the first Brillouin zone of the corresponding spatial and temporal lattices, $k_x \in [-K/2, K/2]$ and $\omega \in [-\Omega/2, \Omega/2]$. Accordingly, the electric field of a mode takes

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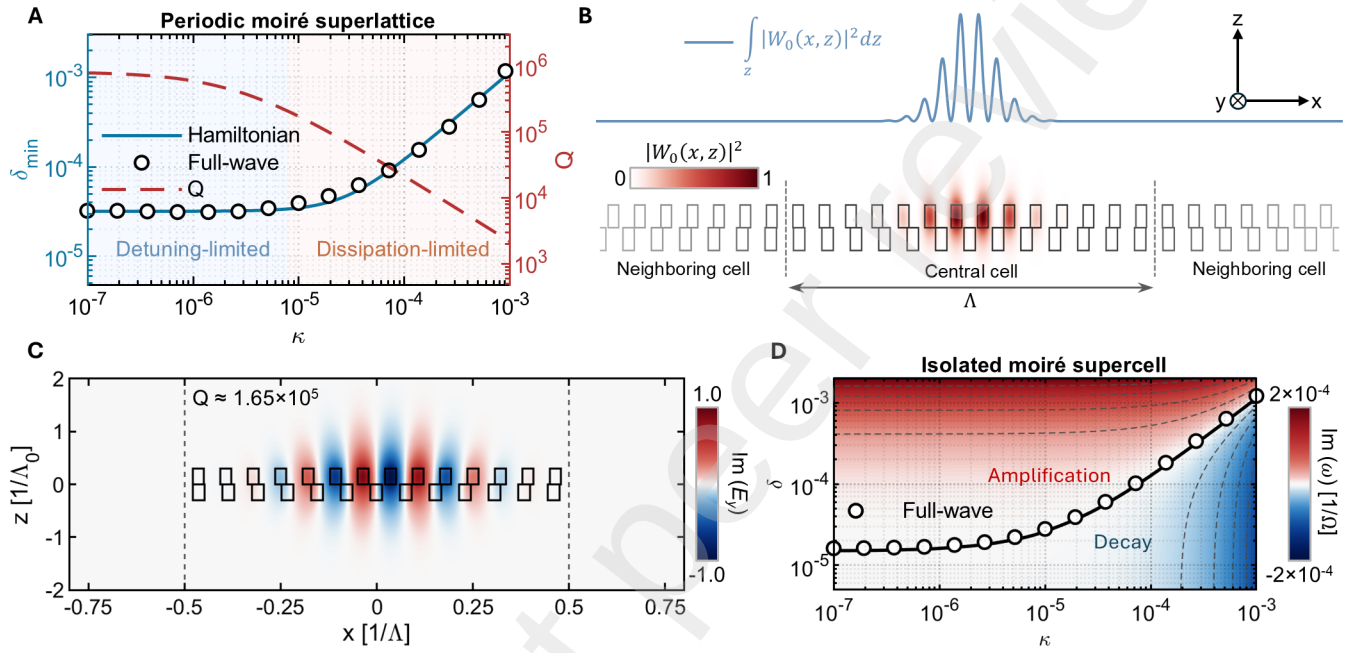


Figure 4: **Material loss and the isolated-supercell limit.** (A) Minimum modulation depth for full-zone amplification as a function of the extinction coefficient κ, shown together with the quality factor. The threshold is dispersion limited while the decay rate stays below the miniband bandwidth, and increases with κ once dissipation dominates. Across the low-loss plateau, the quality factor decreases by nearly an order of magnitude while the threshold remains nearly unchanged, showing that miniband flatness rather than modal lifetime sets the requirement. Symbols denote full-wave results and lines the two-mode Hamiltonian. (B) Wannier density of the flat miniband, with more than 95% confined within a single moiré supercell. (C) Electric-field profile of the corresponding resonance in an isolated supercell, with a quality factor of 1.65×10^5. (D) Amplification threshold of the isolated supercell as a function of κ, using the same notation as in (A). In the absence of miniband dispersion, the criterion reduces to a balance between modulation-induced coupling and decay, recovering the parametric-resonance condition of an open resonator.

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the form

$$\mathbf{E}(x,z,t) = e^{ik_x x} e^{-i\omega t}\, \mathbf{u}_{k_x,\omega}(x,z,t), \tag{6}$$

where $\mathbf{u}_{k_x,\omega}$ is periodic in space and time. Expanding the periodic part as a spatiotemporal Fourier series,

$$\mathbf{u}_{k_x,\omega}(x,z,t) = \sum_{m,n} \mathbf{c}_{mn}(z)\, e^{imKx} e^{-in\Omega t}, \tag{7}$$

the mode is therefore composed of harmonics with wavevectors $k_x + mK$ and frequencies $\omega + n\Omega$, coupled through the spatiotemporal susceptibility of the modulated layer. Matching the harmonic amplitudes across the layers of the structure yields a Floquet–Bloch scattering matrix $\mathbf{S}$ that encapsulates the full spatiotemporal response and relates the incoming and outgoing amplitudes across all Floquet–Bloch channels,

$$\begin{bmatrix} \mathbf{b}^- \\ \mathbf{a}^+ \end{bmatrix} = \mathbf{S}(k_x,\omega) \begin{bmatrix} \mathbf{a}^- \\ \mathbf{b}^+ \end{bmatrix}, \tag{8}$$

where $\mathbf{a}^{\pm}$ and $\mathbf{b}^{\pm}$ collect the forward- and backward-propagating harmonic amplitudes in the regions above and below the structure.

The band structure is identified from the minima of $\sigma_{\min}$, the smallest singular value of $\mathbf{S}^{-1}$, which vanishes at the eigenmodes of the structure, where a nontrivial outgoing solution exists in the absence of an incident field. Quality factors of the static modes and complex eigenfrequencies within the momentum bandgap are obtained by analytically continuing $\sigma_{\min}$ into the complex-frequency plane. For each Floquet–Bloch channel, the Riemann sheet of the longitudinal wavenumber is chosen according to the sign of the corresponding harmonic frequency, ensuring that the outgoing radiation condition is satisfied for both positive- and negative-frequency harmonics. At each mode, the right singular vector of $\mathbf{S}^{-1}$ associated with $\sigma_{\min}$ provides the harmonic amplitudes in the surrounding media, from which the complete internal field distribution is reconstructed throughout the structure.

Numerical implementation

In practice, the spatiotemporal harmonic expansion is truncated by retaining spatial orders $m \in [-N_m, N_m]$ and temporal orders $n \in [-N_n, N_n]$, reducing the scattering problem to a finite-dimensional numerical system that can be solved efficiently. Since each moiré supercell contains N periods of the upper grating and $N-1$ periods of the lower grating, the spatial basis must resolve the finer periodicities of the constituent gratings rather than only the moiré superlattice period, requiring a correspondingly large number of spatial harmonics. Throughout this work, we use $N_m = 200$, for which the flat-band frequency and residual miniband dispersion are converged and agree closely with independent finite-element eigenfrequency calculations performed in COMSOL Multiphysics. For the temporal expansion, $N_n = 1$ is sufficient in the perturbative modulation regime considered here, as verified by comparison with calculations retaining higher temporal orders. The corresponding convergence studies are presented in Section S4 of the Supporting Information.

The isolated-supercell limit is treated within the same periodic scattering formulation by introducing a vacuum spacer between neighboring periodic images of the moiré supercell. The spacer width is increased until the calculated resonance frequency and quality factor become insensitive to further increases, ensuring that coupling between adjacent periodic images is negligible and that the calculated resonance accurately represents the isolated-supercell response.

Declaration of Interests

The authors declare no competing interests.

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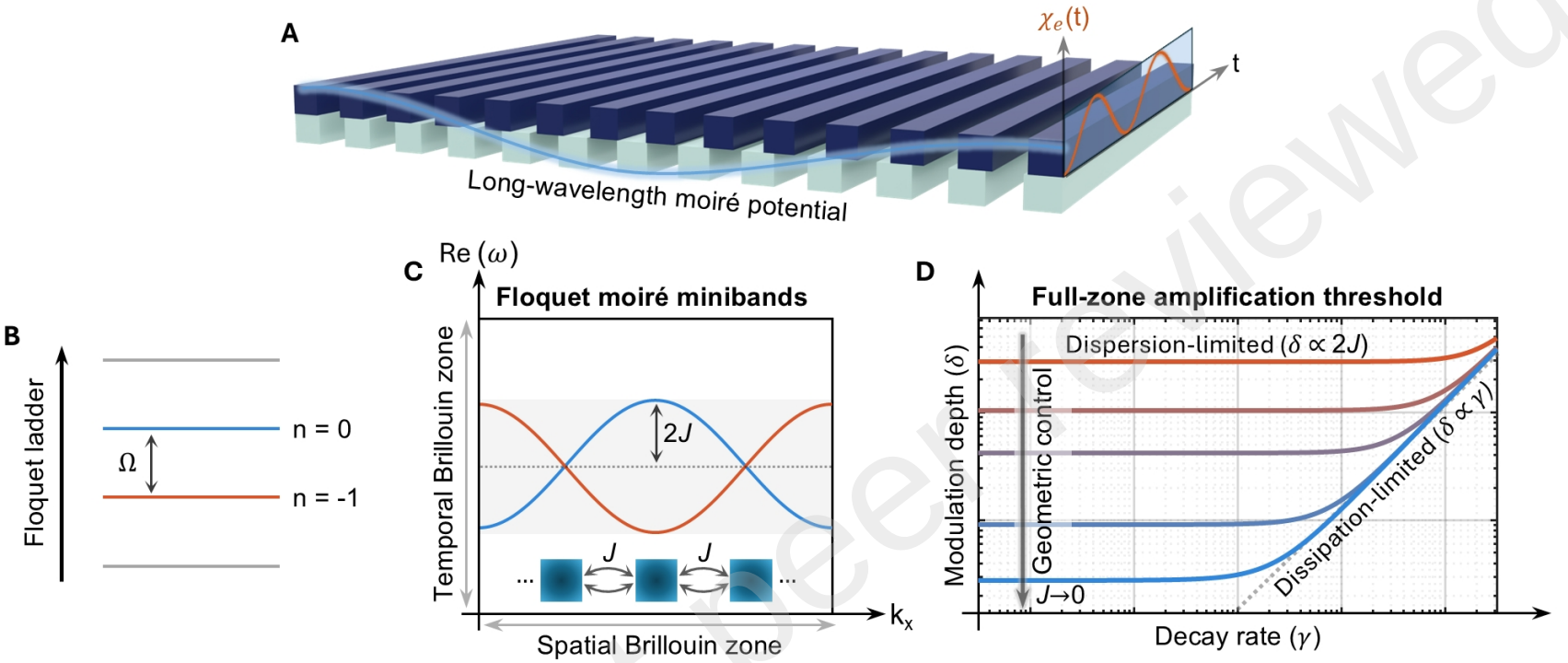
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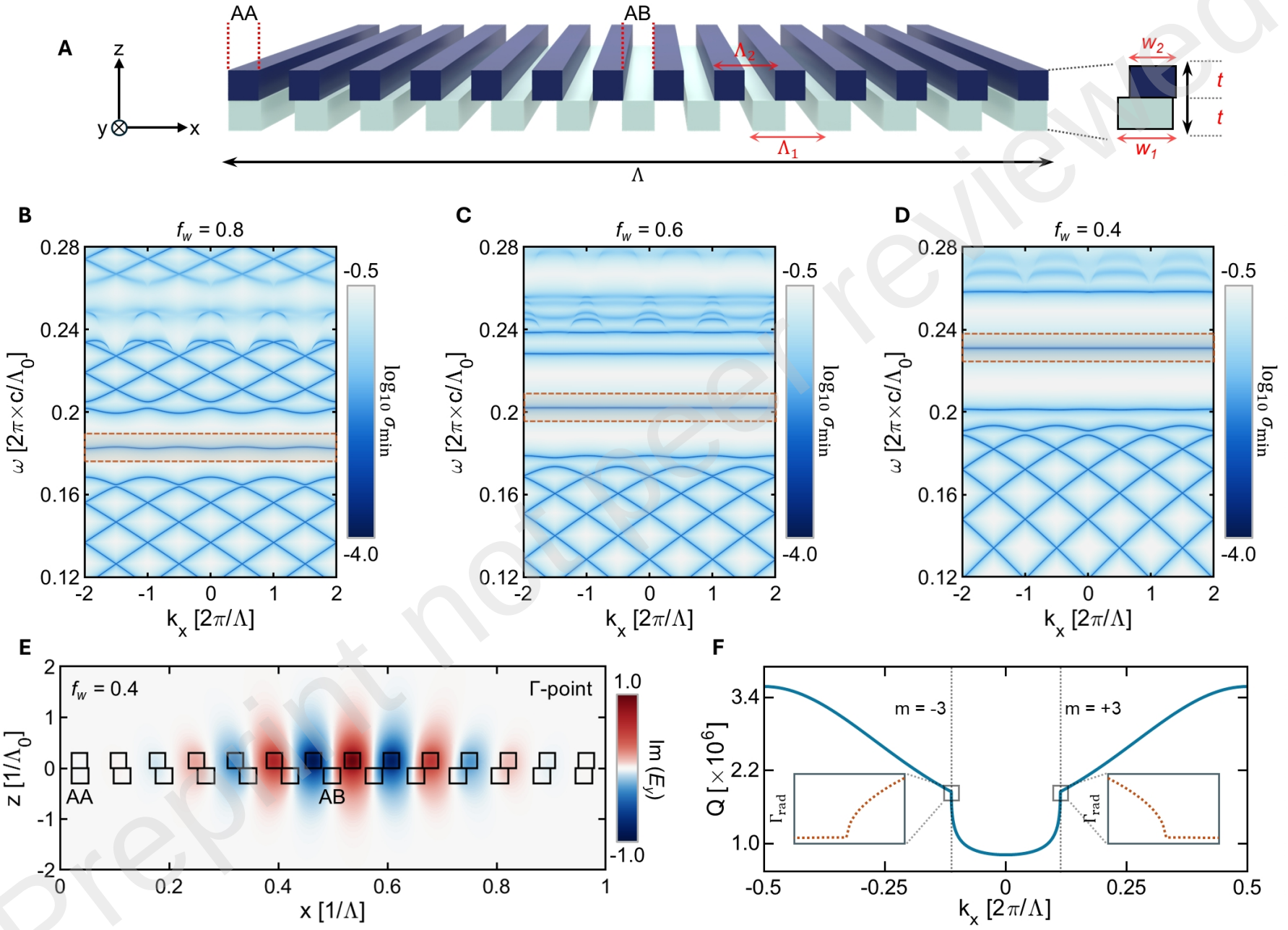
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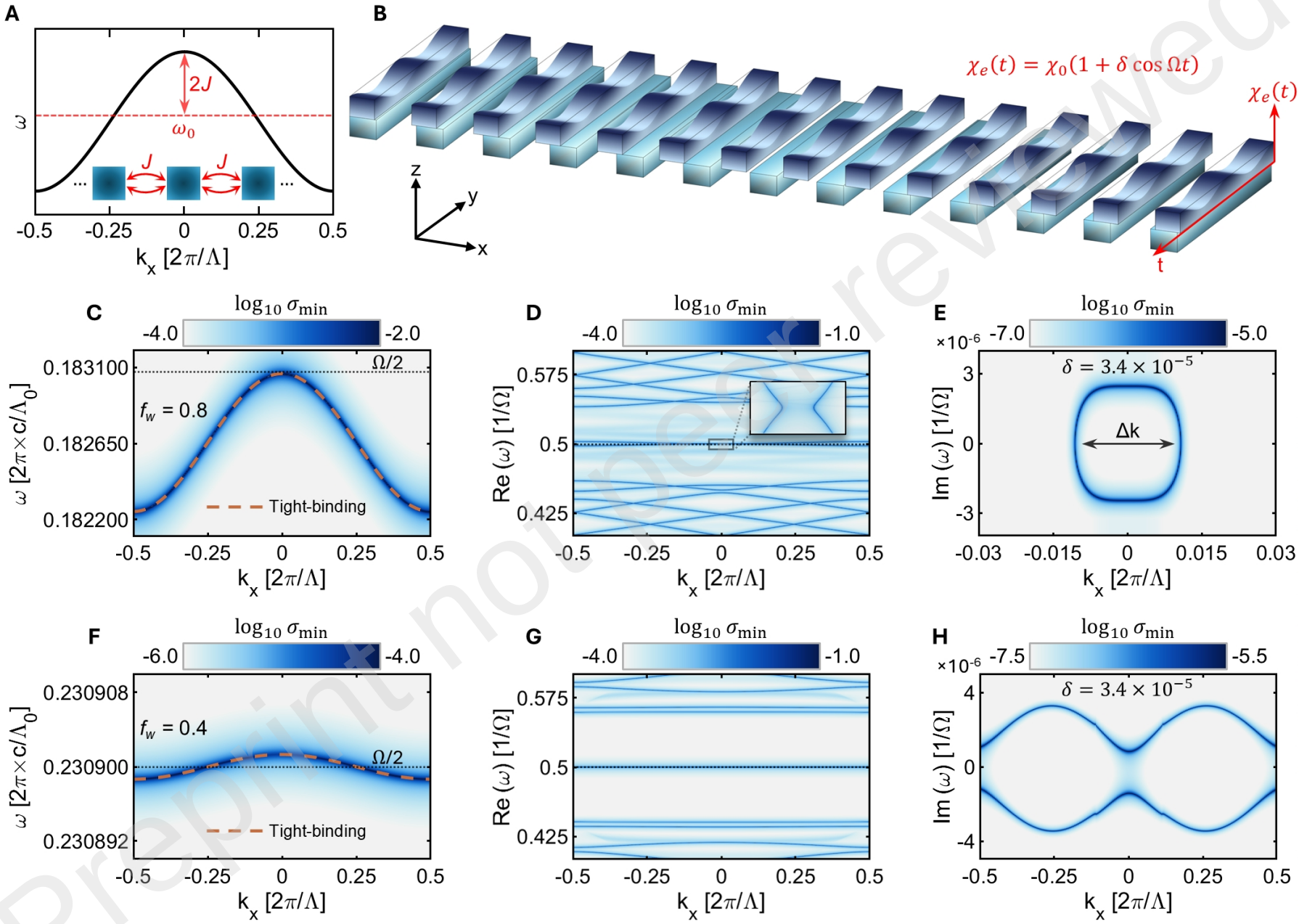
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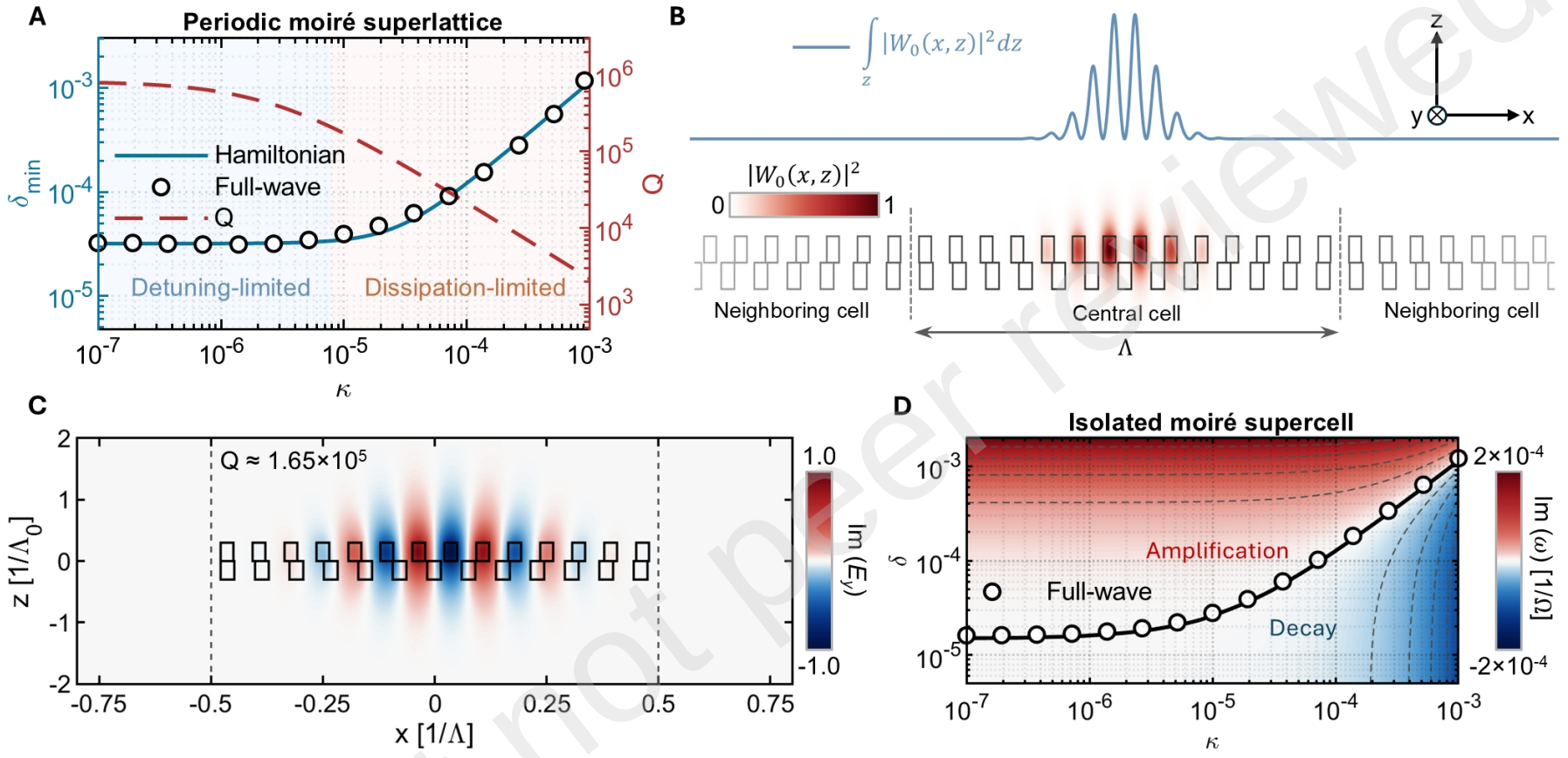
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